

Shubh Pathak

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Summary

AI Engineer at UBS with 2+ years building production ML pipelines and LLM systems for Investment Banking. Currently developing a generative AI compliance-review platform processing 8,000+ documents monthly using Azure OpenAI, hybrid rule-LLM architectures, and async multi-threading. Previously built a machine learning model that drove 4,000+ annual IT infrastructure decisions and ETL pipelines integrating 85+ enterprise systems. Also ships full-stack, production-ready applications end to end, from database schema and security to live deployment, choosing the right tool for each layer.

Experience

- UBS**

Pune, India

Nov 2025 – Present

AI Engineer
 - Led development of a generative AI compliance platform that reduced a 6-person team's manual review workload by 85%, saving 200+ hours monthly, processing 8,000+ Investment Banking documents monthly via a document engine (PDF, PPT, DOCX) with Azure OpenAI vision checks and a commentary engine for emails using grouped LLM prompts
 - Reduced API calls by 67% and compute token cost by 45% by implementing contextual routing that directs document chunks to relevant compliance checks based on task complexity, solving an $M \times N$ scaling problem
 - Engineered async multi-threading via ThreadPoolExecutor for parallel document parsing and model inference, slashing end-to-end pipeline latency by 70%
 - Collaborated with Investment Bank leadership on compliance KRI frameworks and real-time Power BI breach monitoring
 - Migrated 4 of 11 Dataiku apps to RiskLab solo in 3 months before deprecation, turning bare recipes into full user-facing apps with custom UI and backend, cutting costs and lifting performance and user feedback
- Data Analyst and ML Engineer**

Aug 2024 – Oct 2025
 - Streamlined approval and rejection decisions for 4,000+ IT infrastructure tickets annually by engineering a predictive ML model trained on historical virtual server utilization data covering CPU, memory, and disk metrics
 - Slashed hardware approval cycles from 14 days to under 3 days by integrating telemetry from 85+ enterprise systems and deploying rule-based recommendation logic in PostgreSQL
 - Decreased manual data-gathering effort by 40% and accelerated infrastructure assessment cycles by 60% by architecting the INSPIRE dashboard and Solution Designer Studio in Power BI, integrating datasets from 80+ enterprise systems with personalized, reusable visuals
- Data Analyst Intern**

Jan 2024 – Jul 2024
 - Shortened issue resolution time by 85%, from 2-3 days to under 6 hours, via a Power BI failure-monitoring dashboard, and migrated compliance reporting from Tableau to Power BI through orchestrated ETL pipelines into MS SQL

Projects

- InvestIQ – AI Research Platform** | *Next.js, TypeScript, Supabase, Vercel AI SDK* | [Live Demo](#) | [GitHub](#)
 - Collapsed multi-source investment research from hours of manual gathering to a single query by building an agentic AI workspace that plans and runs only the data tools a question needs and returns a structured, sourced report in seconds
 - Enforced database-tier tenant isolation with Postgres Row Level Security validated by a dedicated RBAC test suite, then shipped a tenant-scoped RAG pipeline on pgvector to production on Vercel with a WCAG AA dark-first design system
- Ada – AutoML Agent** | *Python, LangGraph, FastAPI, React, OpenAI API* | [GitHub](#)
 - Architected a 9-agent LangGraph AutoML platform automating the full ML workflow from data profiling to deployment, with a FastAPI and React UI and self-correcting LLM execution with auto-retry on failure

Technical Skills

Machine Learning and AI: Python, TensorFlow, PyTorch, Azure OpenAI, LLMs, NLP, RAG, Agentic AI, LLM Tool-Calling, Prompt Engineering, Structured Output, LangGraph, Model Orchestration, Deep Learning

Full-Stack and Data Engineering: Next.js, TypeScript, Supabase, Vercel, Docker, Azure Data Factory, PostgreSQL, Row Level Security, RBAC, Vector Search (pgvector), Multi-Tenant SaaS, MS SQL, pandas, PyMuPDF, python-docx, python-pptx, openpyxl, Azure, AWS, FastAPI, Git, Power BI, Tableau

Certifications

Microsoft Certified: Azure AI Engineer Associate (AI-102)	Microsoft, 2024
Microsoft Certified: Azure Data Scientist Associate (DP-100)	Microsoft, 2024
TensorFlow Developer Certificate	Google, 2024

Education

Vellore Institute of Technology Bachelor of Technology, Computer Science GPA: 8.70 / 10.00	Bhopal, India Oct 2020 – Oct 2024
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